Body Temperature Regulation and Hypothermia

Guyton and Hall Medical Physiology 14th Edition

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Core Temperature vs. Skin Temperature

- **Core Temperature**
- Temperature of deep tissues (liver, brain, heart, muscles)
- Remains constant: ±1°F (±0.6°C) in healthy individuals
- Normal range:
  - Oral: 97°F - 99.5°F (36°C - 37.5°C)
  - Rectal: ~1°F higher
- Increases with exercise (up to 101°F-104°F)
- **Skin Temperature**
- Varies with environmental temperature
- Critical role in heat loss to surroundings

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Core Temperature vs. Skin Temperature

Parameter	Core Temperature	Skin Temperature
Definition	Temperature of deep tissues	Temperature of body surface
Stability	Constant within ±1°F (± 0.6°C)	Varies with surroundings
Normal Range (Oral)	97°F - 99.5°F (36°C - 37.5°C)	Not fixed
Rectal	~1°F higher than oral	N/A
Exercise Effect	Can rise to 101°F - 104°F	Increases

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Body temperatures under different Conditions

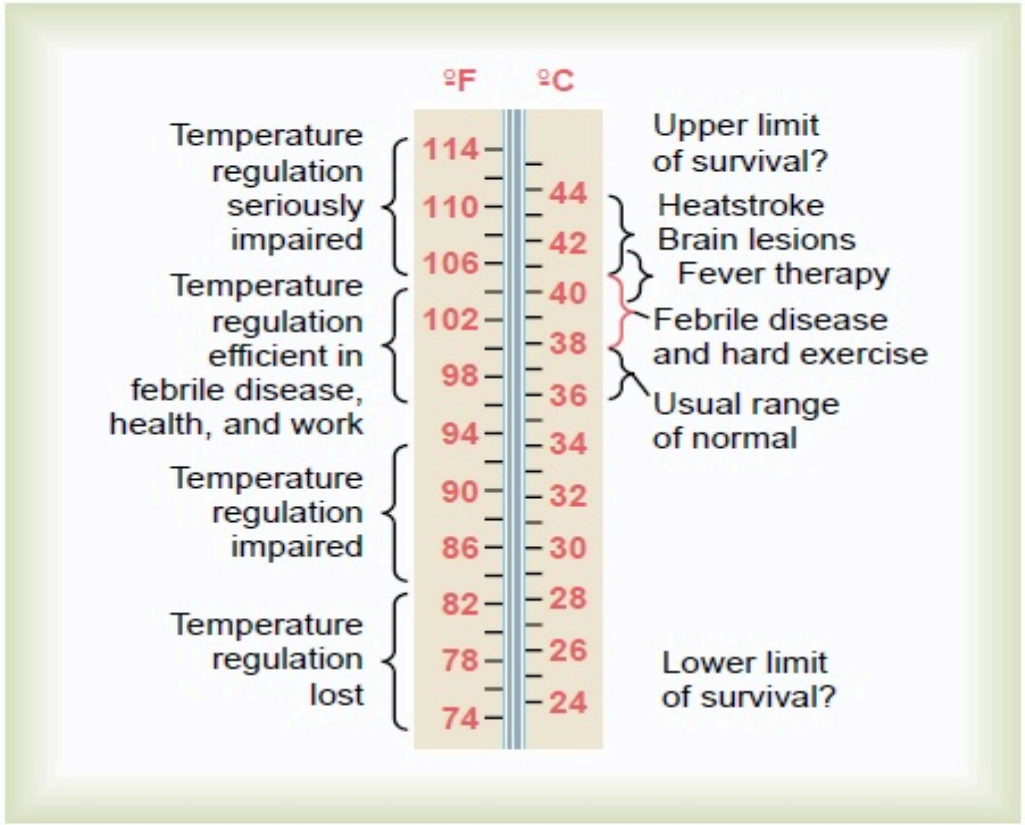

Figure 73–10

Body temperatures under different conditions. (Redrawn from DuBois EF: Fever. Springfield, IL: Charles C Thomas, 1948.)

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Factors Affecting Heat Production

Six Major Determinants of Metabolic Heat Production

- **Basal metabolism** of all body cells
- **Muscle activity** (including shivering)
- **Hormonal effects**: Thyroxine, growth hormone, testosterone
- **Sympathetic stimulation**: Epinephrine, norepinephrine
- **Chemical activity** (increased with rising cell temperature)
- **Thermogenic effect of food** (digestion, absorption, storage)

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Body's Insulator System

- **Components**
- Skin
- Subcutaneous tissues
- **Fat** (conducts heat only **1/3** as readily as other tissues)
- **Insulation Effectiveness**
- **Men:** Equivalent to ¾ of a suit of clothes
- **Women:** Better insulation due to subcutaneous fat
- **Function**
- Maintains internal core temperature while allowing skin temperature to approach environmental temperature

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Blood Flow to Skin Heat Transfer Mechanism

- **Skin Circulation Features**
- Continuous **venous plexus** beneath skin
- **Arteriovenous anastomosis** in hands, feet, ears
- Blood flow can vary from **near zero to 30% of cardiac output**
- Vasoconstricted State → Vasodilated State

1x	8x
(heat conductance)	(heat conductance)

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Methods of Heat Loss from Skin Surface

Method	Description	Typical %
Radiation	Infrared heat rays' emission	60%
Conduction	Direct contact with objects	3% (solids)
Conduction to air	Transfer to air molecules	15%
Evaporation	Sweat vaporization	22%

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Physical Principles of Heat Loss

- **Radiation**
- Electromagnetic waves (5-20 μm wavelength)
- Net loss when body temperature > surroundings
- **Conduction**
- To solids: ~3%
- To air: ~15% (self-limited without convection)
- **Convection**
- Air movement carries heat away
- Cooling effect ∝ √(wind velocity)
- **Water Immersion**
- Heat loss much greater than in air
- No "insulator zone" forms

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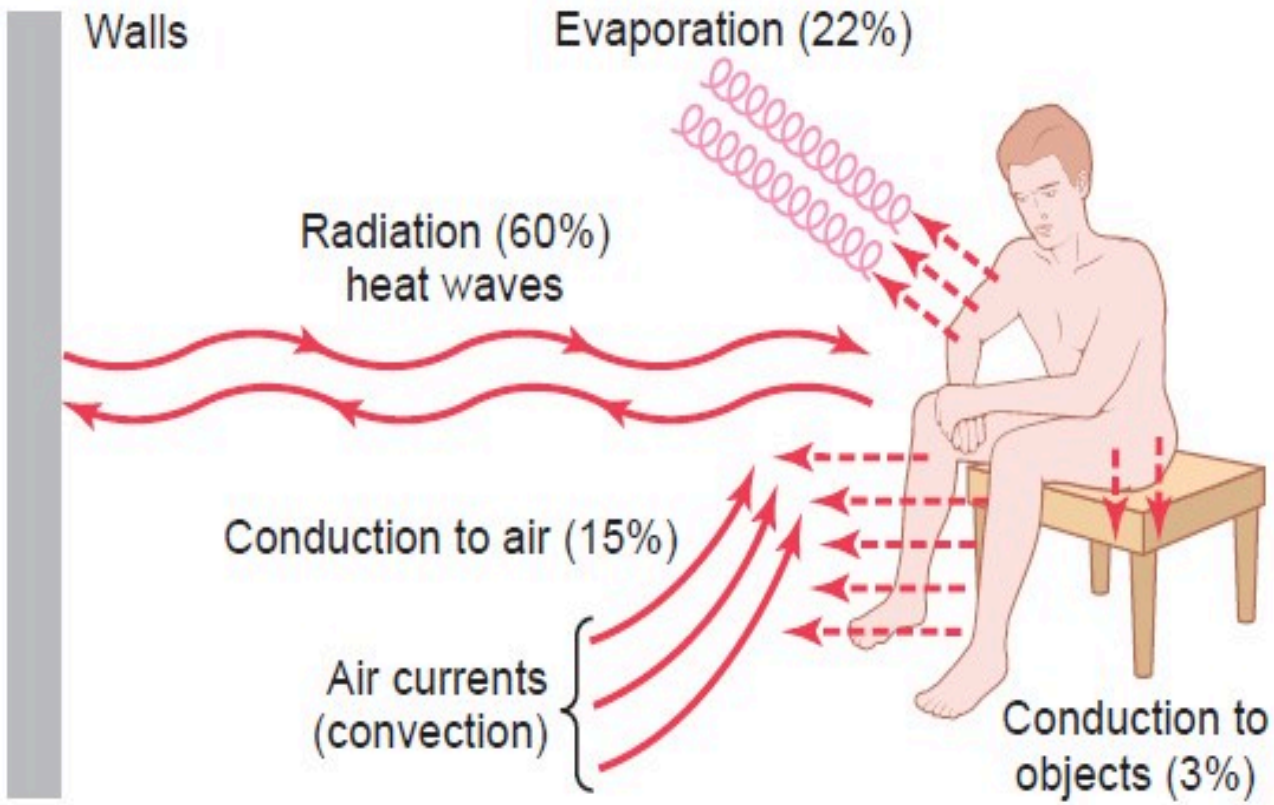

Figure 74-4. Mechanisms of heat loss from the body.

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Evaporation Critical Cooling Mechanism

- **0.58 Calorie** lost per gram of water evaporated
- **Insensible evaporation:** 600-700 mL/day (16-19 Cal/hr.) from skin and lungs
- **Sweat evaporation:** Controllable for temperature regulation
- **When Environmental Temperature > Skin Temperature**
- Radiation and conduction cause **heat GAIN**
- **Evaporation becomes ONLY cooling mechanism**
- **Clinical Note**
- People with congenital absence of sweat glands cannot prevent hyperthermia in hot environments

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Sweating Mechanism

- **Sweat Gland Structure**
- **Coiled secretory portion** (deep subdermal)
- **Duct portion** (through dermis and epidermis)

Sympathetic Cholinergic Nerve

↓

Primary Secretion (protein-free filtrate)

Na+ 142 mEq/L, Cl- 104 mEq/L

↓

Duct Modification (Na+, Cl- reabsorption)

↓

Final Sweat

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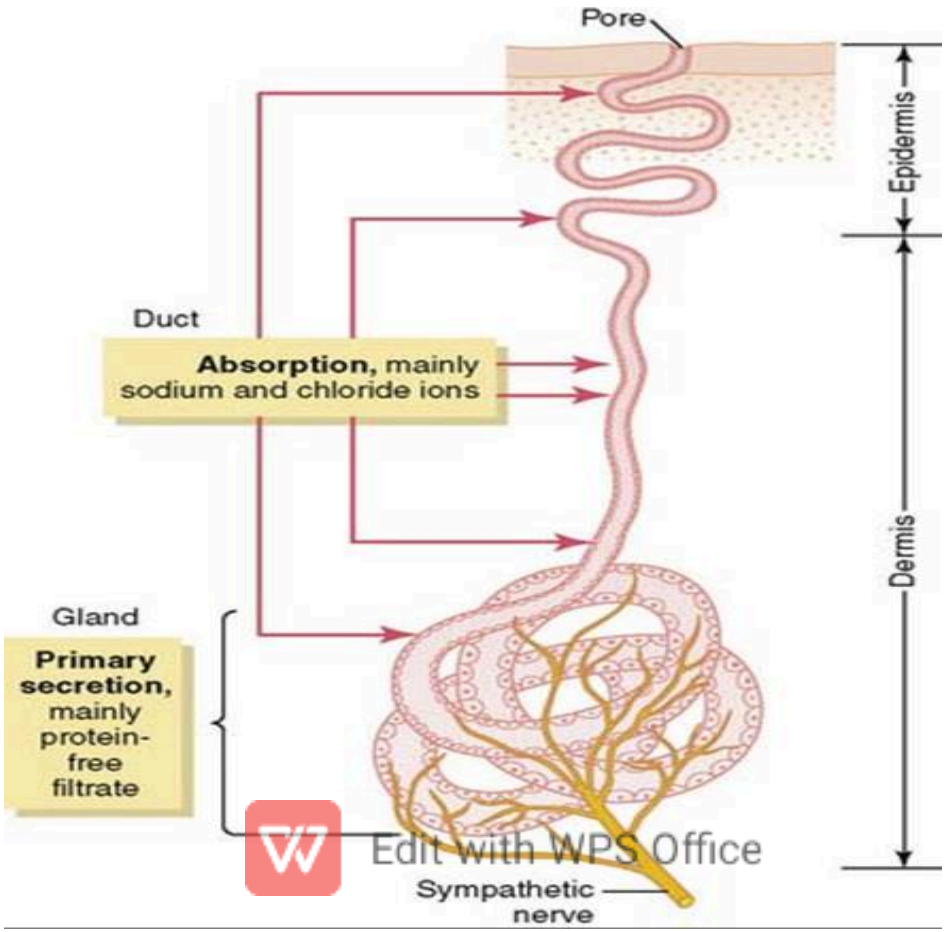
Sweating Mechanism



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Sweat Composition Effect of Sweating Rate

- **Low Sweating Rate**
- Slow duct passage
- **Almost all Na+, Cl- reabsorbed** (as low as 5 mEq/L)
- Water reabsorbed → concentrated urea, lactic acid, K+
- **High Sweating Rate**
- Rapid flow through duct
- **Only 50% Na+, Cl- reabsorbed** (50-60 mEq/L)
- Minimal water reabsorption
- **Acclimatization**
- Max sweat production: **1 L/hr → 2-3 L/hr**
- Na+ loss: **15-30 g/day → 3-5 g/day**
- **Aldosterone** promotes salt conservation

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Hypothalamic Temperature Regulation

- **Anterior Hypothalamic-Preoptic Area**
- Contains **heat-sensitive** and **cold-sensitive neurons**
- Heat-sensitive: 2-10x firing rate with 10°C rise
- Cold-sensitive: Increase firing when temperature falls
- **When Heated:**
- Profuse sweating
- Skin vasodilation
- Inhibition of heat production
- **Conclusion:** Hypothalamic-preoptic area serves as thermostatic control center

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Peripheral Temperature Detection

- **Skin Temperature Receptors**
- **Cold receptors** outnumber **warmth receptors** (10:1)
- Primary function: Detect cool/cold temperatures
- **Transient Receptor protein Channels (TRP channels)** mediate thermal sensation
- **Deep Body Temperature Receptors**
- Located in:
- Spinal cord
- Abdominal viscera
- Great veins (upper abdomen/thorax)
- Detect **cold** primarily
- Function: Prevent **hypothermia**

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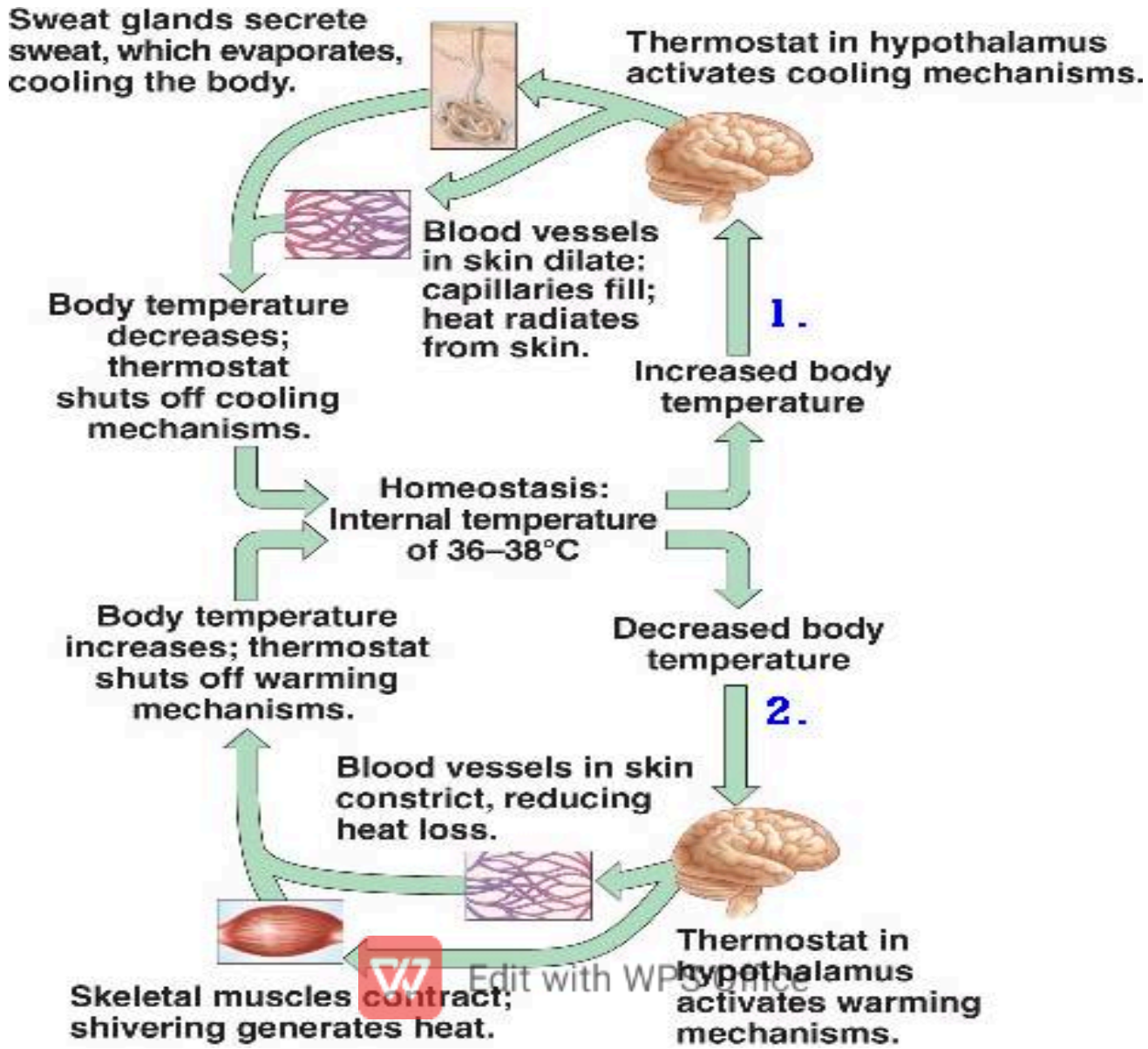

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Reflex Responses to Skin Chilling

- **When Skin is Chilled Throughout Body:**
- **Shivering** ↑ (increases heat production)
- **Sweating inhibited** ↓ (if occurring)
- **Skin vasoconstriction** ↓ (reduces heat loss)
- **Posterior Hypothalamus**
- Integrates signals from:
  - Anterior hypothalamic preoptic area
  - Peripheral receptors
- Controls heat-producing and heat-conserving reactions

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Temperature Decreasing Mechanisms (When Too Hot)

- **Three Important Mechanisms**
- **Vasodilation of skin blood vessels**
  - Inhibits sympathetic vasoconstriction
  - Increases heat transfer to skin **8-fold**
- **Sweating**
  - Sharp increase above **37.1°C (98.8°F)**
  - 1°C rise → 10× basal heat loss
- **Decrease in heat production**
  - Inhibits shivering
  - Inhibits chemical thermogenesis

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Temperature Increasing Mechanisms (When Too Cold)

- **Three Principal Responses**
- **Skin vasoconstriction**
  - Stimulation of posterior hypothalamic sympathetic centers
- **Piloerection** ("goose bumps")
  - Arrector pili muscles contract
  - Traps insulator air (important in animals, minimal in humans)
- **Increase in thermogenesis**
  - Shivering
  - Sympathetic excitation
  - Thyroxine secretion

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Shivering Mechanism

- **Primary Motor Center for Shivering**
- Located in **dorsomedial posterior hypothalamus**
- Normally inhibited by anterior hypothalamus
- Excited by cold signals from skin and spinal cord
- **Shivering Process**
- Increased skeletal muscle tone
- Feedback oscillation of muscle spindle stretch reflex
- Rhythmic shaking begins
- **Heat Production**
- Maximum shivering → **4-5× normal** heat production

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Chemical (No shivering) Thermogenesis

- **Sympathetic Stimulation**
- Epinephrine and norepinephrine
- Uncouple oxidative phosphorylation
- Heat released without ATP formation
- **Brown Fat**
- Rich in mitochondria with **thermogenin**
- Rich sympathetic innervation
- **Infants:** Can increase heat production 100%
- **Adults:** Only 10-15% increase (minimal brown fat)

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Thyroxine in Long-Term Temperature Regulation

Cold exposure

↓

Hypothalamus ↑ TRH

↓

Anterior Pituitary ↑ TSH

↓

Thyroid Gland ↑ Thyroxine

↓

Uncoupling protein activation

↓

↑ Cellular metabolism

Edraw Max Trial

Thyroxine in Long-Term Temperature Regulation

Step	Gland/Structure	Hormone/Substance	Full Form	Function
1	Hypothalamus	**TRH**	**Thyrotropin-Releasing Hormone**	Stimulates anterior pituitary
2	Anterior Pituitary	**TSH**	**Thyroid-Stimulating Hormone** (Thyrotropin)	Stimulates thyroid gland
3	Thyroid Gland	**T4**	**Tetraiodothyronine** (Thyroxine)	Prohormone (converted to T3)
4	Peripheral Tissues	**T3**	**Triiodothyronine**	Active hormone (3-5 × more potent than T4)
5	Brown Adipose Tissue	**UCP1**	**Uncoupling Protein 1** (Thermogenin)	Generates heat by uncoupling oxidative phosphorylation
6	All Cells			**↑ Cellular metabolic rate**

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Thyroxine in Long-Term Temperature Regulation

- **Time Course**
- Requires **weeks** of cold exposure
- Thyroid can enlarge 20-40%
- **Clinical Note**
- Higher incidence of toxic goiters in cold climates

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Temperature "Set Point"

- Critical body core temperature (~**37.1°C / 98.8°F**) where:
- Above: Heat loss > Heat production
- Below: Heat production > Heat loss
- **Feedback Gain**
- Body temperature changes **1°C** per **25-30°C** environmental change
- **Feedback gain ≈ 27** (extremely high for biological system)
- Compare: Baroreceptor system gain < 2

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Skin Temperature Effect on Set Point

- **Effect on Sweating Set Point**

Skin Temperature	Hypothalamic Set Point for Sweating
>33°C	36.7°C
29°C	37.4°C

- **Effect on Shivering Set Point**
- Cold skin drives shivering threshold **even when hypothalamic temperature is normal**
- **Physiological Value:** "Anticipates" fall in core temperature

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Behavioral Temperature Control

- **Conscious Regulation**
- More **powerful** than subconscious mechanisms
- Triggered by psychic sensations:
  - Overheated discomfort
  - Cold discomfort
- **Examples**
- Moving to heated/cooled environment
- Adjusting clothing
- Seeking shade or sun
- **Clinical Significance**
- **Only effective mechanism** in severely cold environments
- Essential for spinal cord injury patients

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Fever
Definition and Causes

- Body temperature above normal range
- **Causes**
- **Brain abnormalities**
  - Tumors compressing hypothalamus
  - Surgical trauma to hypothalamus
- **Toxic substances (Pyrogens)**
  - Bacterial infections (endotoxins from gram-negative bacteria)
  - Viral infections
  - Tissue breakdown products
- **Environmental conditions**
  - Heatstroke

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Pyrogens and Fever Mechanism

- **Pyrogens**
- Substances that raise hypothalamic set point

Bacteria/tissue breakdown

↓

Phagocytized by leukocytes, macrophages, killer lymphocytes

↓

Release of CYTOKINES (especially Interleukin-1 / IL-1)

↓

Hypothalamus

↓

Prostaglandin E2 formation

↓

↑Set point

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Pyrogens and Fever Mechanism

- IL-1 = leukocyte pyrogen = endogenous pyrogen
- Nanogram quantities sufficient
- **Aspirin** reduces fever by blocking prostaglandin formation

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Febrile Response Chills and Crisis

- **Phase 1: Set Point Increased (Chills)**
- Vasoconstriction
- Piloerection
- Epinephrine secretion
- Shivering
- Patient feels cold despite rising temperature
- **Phase 2: Temperature at New Set Point**
- No chills
- Neither hot nor cold sensation
- Temperature regulated at higher level
- **Phase 3: Set Point Decreased (Crisis/"Flush")**
- Intense sweating
- Vasodilation (hot skin)
- Temperature falls to normal

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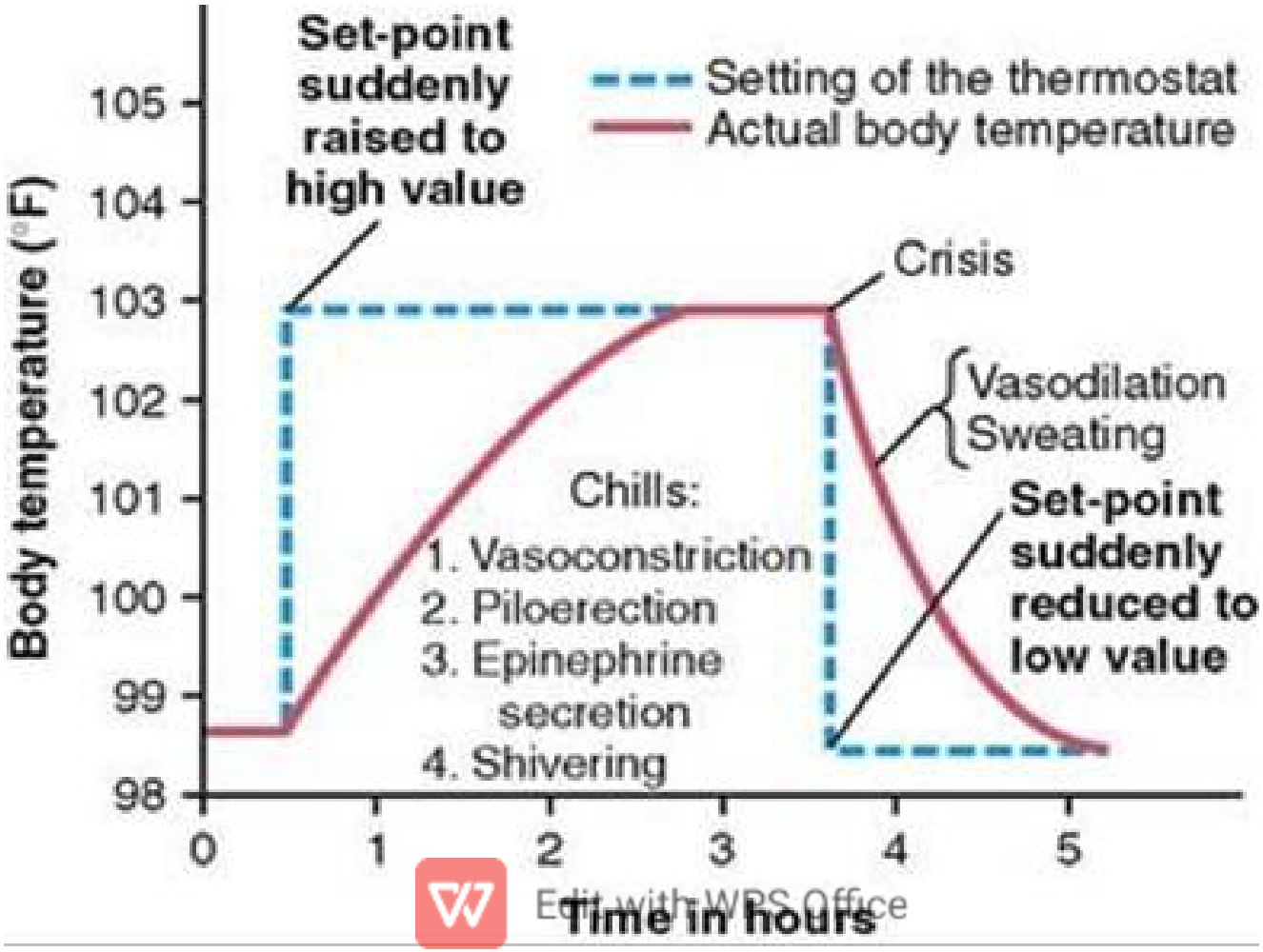
Fever Curve



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Heatstroke

Critical temperature: 105°F - 108°F

- High humidity + high temperature = greatest risk
- Heavy work in 85°F - 90°F can trigger heatstroke
- **Symptoms**
- Dizziness
- Abdominal distress, vomiting
- Delirium
- Loss of consciousness
- Circulatory shock (fluid/electrolyte loss)
- **Treatment**
- Immediate cooling (cold water bath, spray cooling)
- Monitor for shivering (increases heat production)
- Muscle relaxants if needed

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Heat Acclimatization

- **Time Course**
- 1-3 weeks of daily heat exposure + workload
- **Physiological Changes**

Parameter	Before	After
Max sweat rate	~1 L/hr	2-3 L/hr
Salt loss	15-30 g/day	3-5 g/day
Plasma volume	Baseline	Increased
Aldosterone	Normal	Increased

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Cold Exposure Hypothermia

- **Critical Thresholds**
- **Loss of temperature regulation:** Below 94°F (impaired)
- **Hypothalamic control lost:** Below 85°F
- **Lethal (ice water):** 20-30 minutes → core temp ~77°F
- **Causes of Death**
- Heart standstill
- Ventricular fibrillation
- **Mechanisms of Failure**
- Cellular metabolism ↓ 2× per 10°F drop
- CNS depression (sleepiness → coma)
- Shivering ceases

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Frostbite and Cold Protection

- **Frostbite**
- Affects: Ear lobes, digits (hands, feet)
- Ice crystal formation in cells
- Permanent damage:
  - Circulatory impairment
  - Tissue damage
  - Gangrene (may require amputation)
- **Cold-Induced Vasodilation (Protective)**
- At near-freezing temperatures
- Vascular smooth muscle paralysis
- Sudden vasodilation (skin flush)
- Delivers warm blood to skin
- **Less developed in humans** than cold-adapted animals

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Artificial Hypothermia

- **Induction Method**
- Strong sedative (depresses hypothalamic controller)
- Cooling with ice or cooling blankets
- Maintain <90°F with cool water/alcohol spray
- **Duration**
- Days to weeks
- **Clinical Application: Cardiac Surgery**
- Heart can be stopped for **many minutes**
- Cell metabolism greatly depressed
- Cells survive **30-60 minutes** without blood flow
- **Safety**
- No tissue damage at these temperatures

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Summary
Temperature Regulation System

The hypothalamic temperature control center maintains a set point of 37.1°C (98.8°F) by integrating input from three types of temperature receptors: skin receptors (with a 10:1 ratio of cold to warmth receptors), deep body receptors (located in the spinal cord, viscera and great veins) and hypothalamic receptors themselves. In response to these signals, the hypothalamus orchestrates a balance between heat production mechanisms (basal metabolism, muscle activity, hormones and shivering) and heat loss mechanisms (radiation, conduction, convection and evaporation) to maintain core temperature stability.

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Summary

The temperature of the body is regulated almost entirely by nervous feedback mechanisms and almost all these mechanisms operate through temperature-regulating centers located in the hypothalamus.

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Thank You